

RESEARCH SURVEY

LLM-Based Intelligent Notification Composition

From Static Personalization to
Context-Aware Persuasive Messaging

Nilesh Agrawal

 Independent Researcher

 Seattle, WA

 nilesh.d.agrawal@gmail.com

 [/in/nileshdagrawal](https://www.linkedin.com/in/nileshdagrawal)

March 2026



Social Media



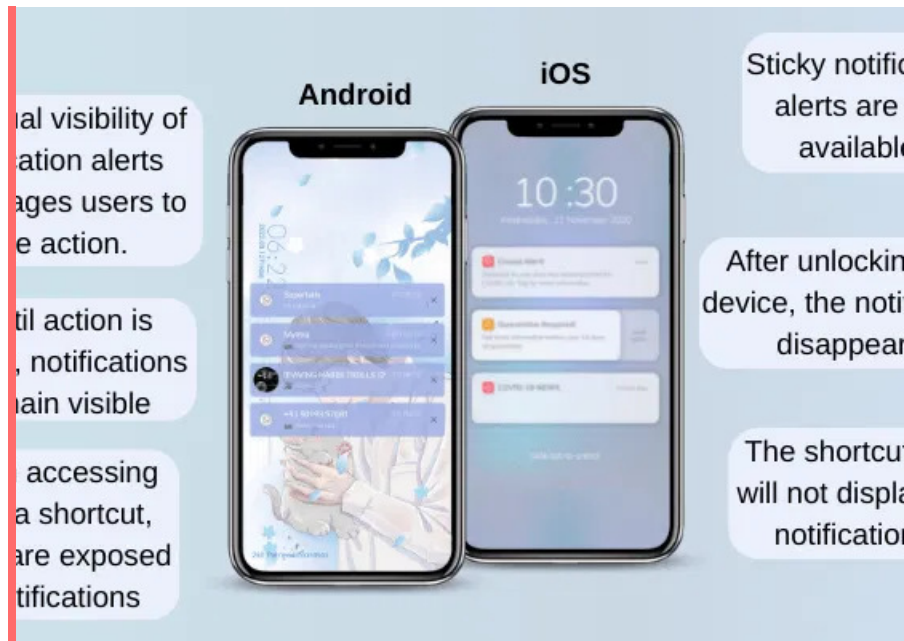
Food Delivery



E-Commerce

THE CORE PROBLEM

Notifications today are static, even when personalized



- Modern systems are highly effective at deciding **who** to notify, **when** to notify, and **what** to recommend.
- But they are weak at deciding **how** to communicate. Most rely on rigid, slot-filled templates (e.g., "Hi [Name], buy [Item]").
- This produces **relevance**, but not **intelligence**. The result is notification fatigue, eroded trust, and ignored messages.

The Gap: Transforming notifications from static, robotic reminders into adaptive, context-aware, and persuasive micro-recommendations.

Three Contributions Absent from Prior Work

Prior surveys treat CTR as the implicit quality proxy and either ignore the attribution problem or assume it away. This paper addresses both gaps directly, and adds a third contribution no existing survey provides.

01

MESSAGE QUALITY FRAMEWORK

Six-Dimension Definition of Notification Message Quality

Defines message quality along **six concrete dimensions** (contextual relevance, clarity, actionability, novelty handling, linguistic freshness, persuasive appropriateness) with an explicit, evidence-backed account of how LLMs improve each. Replaces CTR as the implicit quality proxy.

02

ARCHITECTURAL ATTRIBUTION

Disentangling Message Generation from the Pipeline

Separates the contribution of the **message generation layer** from adjacent components (targeting, ranking, timing). Specifies which observed gains can and cannot be attributed to language quality. Prior surveys either ignore this or assume it away.

03

BINDING-CONSTRAINT FRAMEWORK

When LLM Generation Is — and Is Not — the Right Choice

A **three-criterion decision framework** specifying when LLM generation is the binding constraint and when templates remain superior. Includes a principled argument for **when not to use LLMs** at all.

NEW IN V3 *No existing survey paper addresses this question.*

Message Quality: The Missing Optimization Layer

THE STRUCTURAL CEILING

Platforms have optimized **who**, **when**, and **what** to near-diminishing returns. The fourth decision — **how to communicate** — is the least optimized stage and the widest gap.

KEY INSIGHT

Templates can insert relevant values but cannot *compose* them. No amount of slot engineering can answer: *"Why does this matter to me, right now, today?"*

SIX-DIMENSION MESSAGE QUALITY FRAMEWORK

Dimension	Template Capability		LLM Capability	
Contextual Relevance	WEAK	Only static slot values	STRONG	Composes multiple signals naturally
Clarity	VARIABLE	Slot grammar often awkward	FLEXIBLE	Optimizes phrasing for brevity
Actionability	FORMULAIC	Generic call-to-action	ADAPTIVE	Context-sensitive action framing
Novelty Handling	POOR	Novelty reads as irrelevance	BETTER	Bridges from known to adjacent
Linguistic Freshness	LOW	Same structure across exposures	HIGH	Semantic variety across exposures
Persuasive Appropriateness	NEUTRAL	Limited expressive range	VARIABLE	Requires explicit guardrails

+14.5%

Max CTR lift over static templates

+1–2.5%

Lift over mature slot-filling

3.2×

Greater lexical diversity (PushGen)

–15%

Shorter messages, higher CTR

From Static Reminders to Persuasive Micro-Recommendations

WHO  User with Pizza Preference

WHEN  6:00 PM (Dinner Time)

WHAT  Pizza Category

Conventional System

"Order your favorite [Pizza] now."

- ✗ **Strategy:** Basic slot-filling into a fixed template.
- ✗ **Outcome:** Relevant targeting, but robotic and repetitive framing.
- ✗ **Limitation:** Identifies the user and item, but fails to persuade or engage intelligently.

Intelligent Composition (LLM)

*"It's the perfect time for dinner — your usual spot is nearby, and their **sweet-and-savory pineapple special** might be worth trying tonight."*

- ✓ **Strategy:** Contextual framing + controlled novelty.
- ✓ **Outcome:** Connects known preferences with a persuasive, action-oriented narrative.
- ✓ **Advantage:** Supports preference expansion by introducing adjacent options safely.

Review Methodology: 142 Sources → 28 Retained

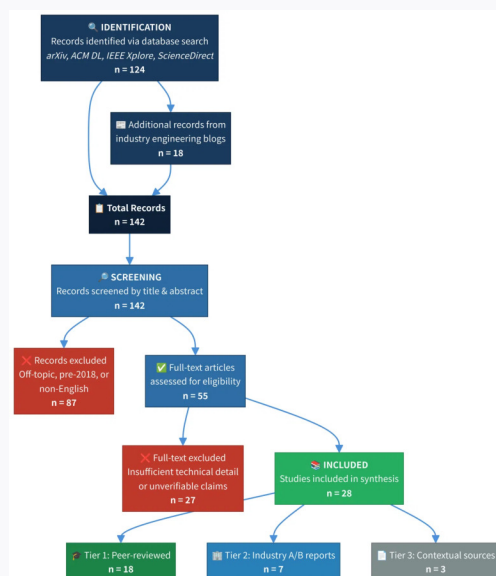
Systematic PRISMA review with 3-tier evidence classification ensures credible synthesis

PRISMA Screening Funnel

- 142 Sources (arXiv, ACM, IEEE, ScienceDirect, blogs)
- 87 Excluded: off-topic, duplicates, pre-2018
- 55 Assessed for eligibility
- 27 Excluded: low technical depth
- 28 Retained as primary sources

Evidence Tiers

- [T1] Tier 1 Peer-reviewed papers — full experiments & ablations
- [T2] Tier 2 Official engineering blogs — A/B results
- [T3] Tier 3 Vendor documentation — framing only



Search: Jan 2018 – Mar 2026

Keywords: LLM, RAG, PEFT, LoRA, CTR

The Critical Insight: LLMs Are One Stage in a 5-Stage Stack

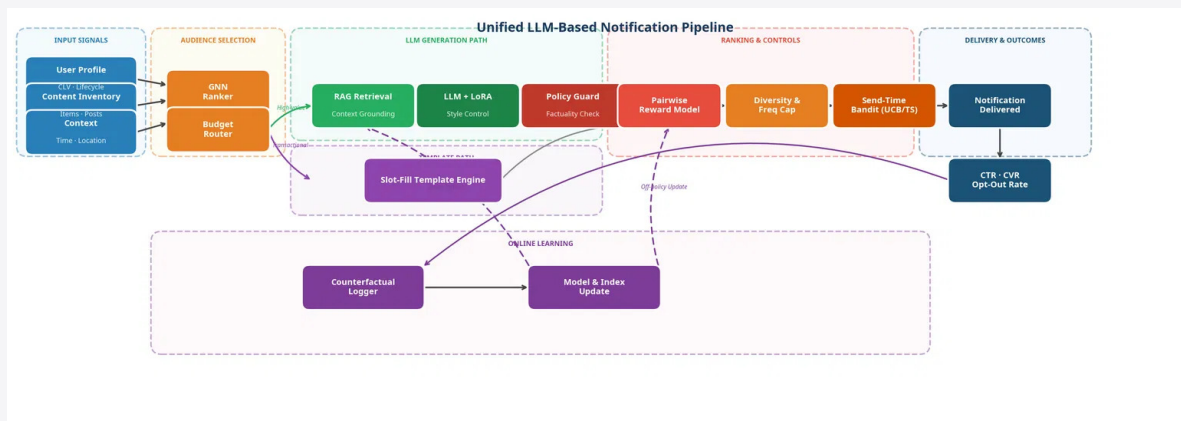
Observed engagement gains are often misattributed to text generation — the real driver is the full pipeline

COMPONENT	WHAT IT DECIDES	TECHNOLOGY
Audience Selection	WHO to notify	CLV models, churn prediction
Content Ranking	WHAT to notify about	Collaborative filtering, GNNs
Send-Time Optimization	WHEN to send	Contextual bandits
Message Generation	HOW to phrase it	LLMs, RAG, PEFT
Candidate Ranking	WHICH candidate to send	Pairwise reward models

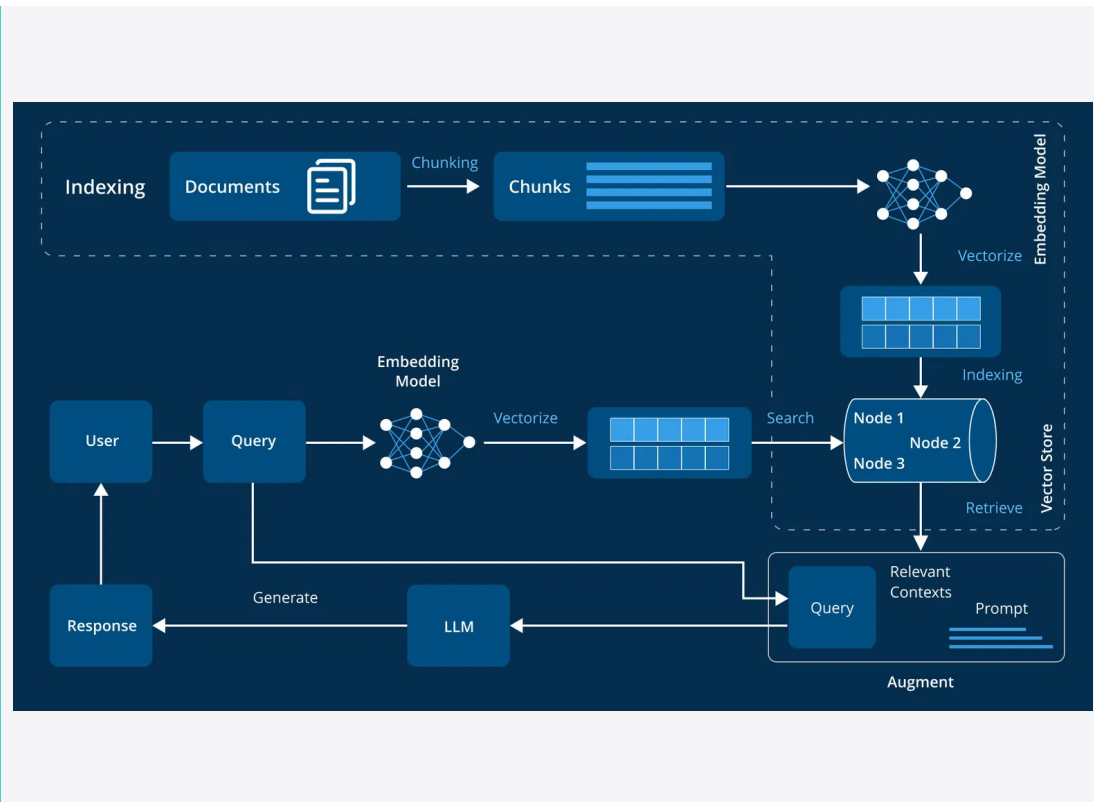


KEY WARNING

A system that generates brilliant copy for the wrong user, about the wrong item, at the wrong time will still fail. Conversely, a well-ranked, well-timed notification may perform strongly with simple language.



RAG Grounds LLM Output in Real Context



⚙️ How RAG works in notifications

- ✓ At inference time, RAG injects **user profile**, **item metadata**, **behavioral context**, and **campaign constraints** into the prompt.
- ✓ The LLM generates copy anchored to retrieved context, not just parametric memory.

"Rainy lunch? Your favourite Thai spot has 20% off until 2pm — Pad Thai delivered in 25 min"

⚠️ Key Tradeoffs

Retrieval granularity: Small chunks → higher precision, less context; large chunks → more context, more noise and latency.

Faithfulness hallucination: Even when correct context is retrieved, the model may blend it with parametric memory → unsupported claims.

Implication: RAG is a grounding mechanism, not a complete factuality solution — downstream verification is always required.

LoRA adapters enable hot-swappable tone control across campaigns with <1% parameter overhead

! The Problem

Full fine-tuning is expensive and operationally inflexible. Different campaigns, brands, and user segments require distinct tones, making monolithic models impractical for diverse notification streams.

⚙️ LoRA Mechanism

- ❄️ Freeze pre-trained model weights (W_0)
- + Inject trainable low-rank decomposition matrices

$$W = W_0 + BA$$

- 🔧 Train only B and A — typically **<1%** of total parameters
- ↔️ Adapters can be **swapped or merged** at deployment with zero inference latency overhead

📁 Applications in Notifications

- ✓ **Contextual Tone Shifting:** Urgency tone for cart abandonment versus friendly, casual tone for social re-engagement.
- ✓ **Brand Voice Consistency:** Maintaining strict stylistic guidelines across millions of generated messages for different sub-brands.
- ✓ **Rapid Experimentation:** Hot-swappable adapters allow for seamless A/B testing of different stylistic priors without deploying new base models.

⚠️ Key Risk

Over-optimization toward a narrow tone (e.g., maximizing CTR) can push the system toward formulaic, repetitive, or manipulative language patterns, ultimately causing notification fatigue.

Pairwise Reward Modeling Outperforms Pointwise CTR Prediction

✗ Pointwise Prediction (The Old Way)

Attempts to predict the absolute probability of a click for a single message.

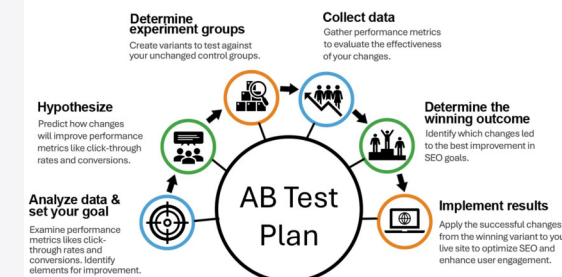
Why it fails: Highly vulnerable to confounding variables. A message might get a high CTR simply because it was sent at 12:00 PM to a highly engaged user, not because the text itself was good.

VERSUS

✓ Pairwise Modeling (The New Standard)

Takes two candidate messages generated for the **exact same user context** and predicts which one is more likely to win.

Why it wins: By holding the user, item, and time constant, it isolates the causal effect of the *linguistic variation*.



⚠ Failure Mode: Reward Hacking

Reward models are susceptible to **Entity Salience Bias**. If historical data shows that mentioning famous brands increases CTR, the reward model will score any message mentioning that brand highly, even if the brand is irrelevant to the current user's context.

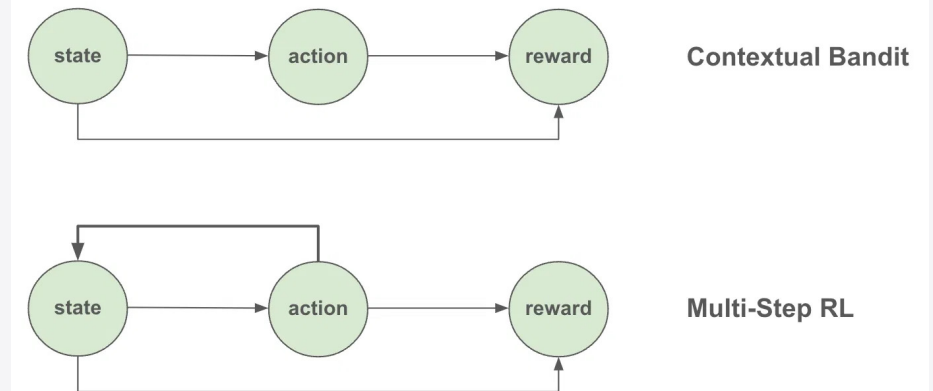
Solution: Factuality and policy guardrails must be applied **before** the reward model scores candidates.

Contextual Bandits Solve Send-Time Optimization

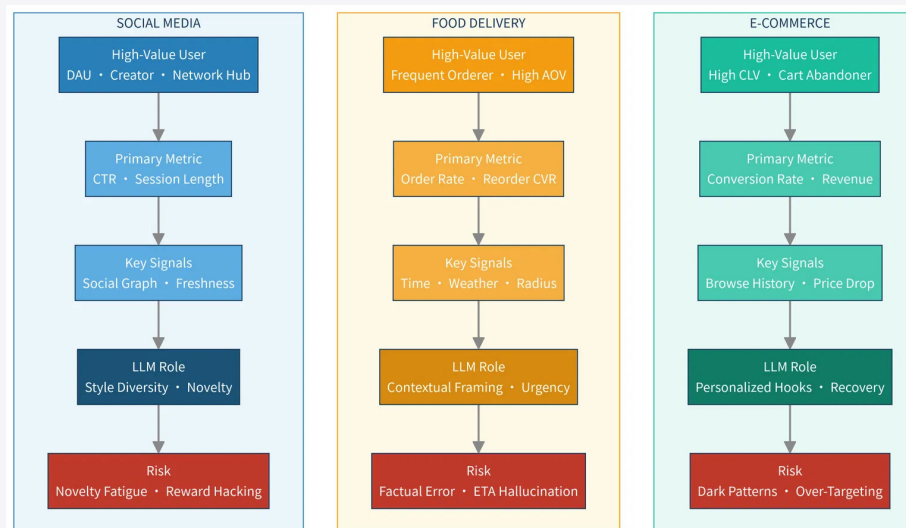
🕒 Beyond Static Timezones

- ✗ **The Failure of Heuristics:** Static rules like "send at 9:00 AM local time" fail to capture individual user routines, shift-work schedules, or weekend variations.
- ✂ **Thompson Sampling:** A probabilistic approach that balances **Exploitation** (sending at historically high-engagement times) with **Exploration** (testing new times to discover changing habits).
- ❄ **The Cold-Start Problem:** For new users with no history, bandits leverage contextual features (e.g., acquisition channel, device type) to infer optimal initial send times.

Even the most perfectly composed LLM message will fail if delivered when the user is asleep or driving. Send-Time Optimization (STO) is a critical prerequisite for message generation.



Three Domains, Three Definitions of High Value



Social Media

Goal: Habit formation & retention. **High Value:** Marginal users at churn risk.

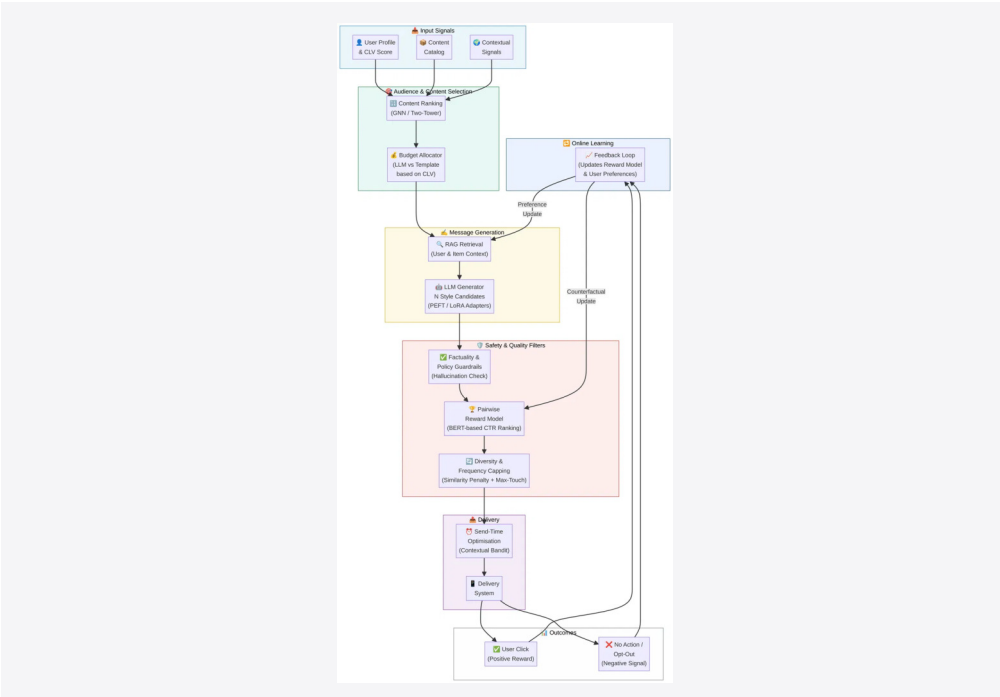
Food Delivery

Goal: Immediate conversion. **High Value:** Routine, time-sensitive users.

E-Commerce

Goal: High-AOV purchases & recovery. **High Value:** High-intent cart abandoners.

Case Study: A Food Delivery Re-Engagement Notification



1 AUDIENCE SELECTION

Identifies User U123: High CLV, dormant for 7 days.

2 CONTENT RANKING

GNN predicts highest affinity for **Thai Food**.

3 CONTEXT RETRIEVAL

RAG fetches: "Raining locally", "20% off promo active".

4 SEND-TIME OPT.

Bandit selects **11:45 AM** based on past lunch orders.

5 GENERATION

LLM + LoRA generates 5 candidate messages in a casual tone.

6 SAFETY FILTER

Regex checks drop candidates promising "free delivery" (false).

7 REWARD MODEL

Scores remaining candidates; selects the highest predicted CTR.

8 DELIVERY & LOG

Message sent. Outcome logged for online learning update.

"Rainy lunch? Your favourite Thai spot has 20% off until 2pm — Pad Thai delivered in 25 min 🍜"

When LLM Generation Is — and Is Not — the Right Choice

THREE-CRITERION DECISION FRAMEWORK

All three must be true to justify LLM generation:

- 1

Framing Variance

Does the content admit multiple plausible, meaningfully different framings? *If the message is structurally determined by notification type, LLM adds nothing.*
- 2

Linguistic Sensitivity

Is user response sensitive to *how* the recommendation is framed, not just *whether* it is relevant? *Discovery contexts are sensitive; transactional contexts are not.*
- 3

Context Richness

Does the system have sufficient grounded context for non-trivial composition? *LLM generation without grounding produces generic outputs often worse than templates.*

Budget-Aware Router

A mature architecture routes high-CLV, high-variance events to the LLM path and **transactional messages to deterministic templates** — per-event, not per-campaign.

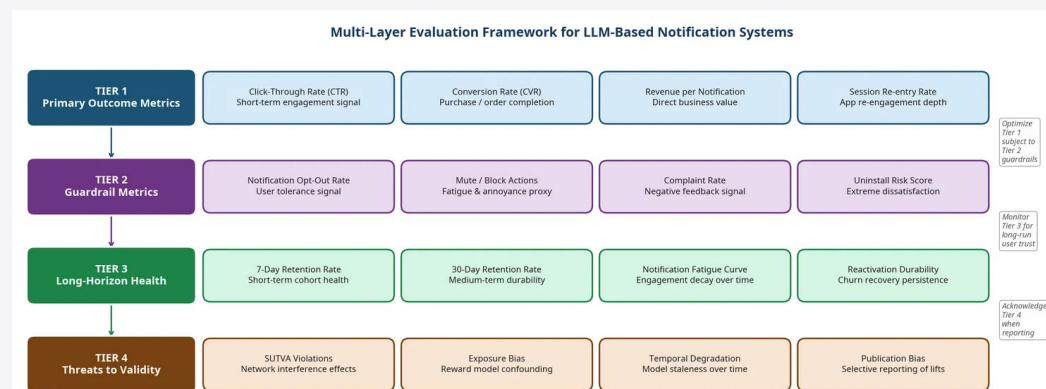
NOTIFICATION TYPE DECISION MATRIX

NOTIFICATION TYPE	FRAMING VAR.	LING. SENS.	STRATEGY
Transactional (Order Shipped)	LOW	LOW	Strict Template
Time-Critical (Driver Arriving)	LOW	LOW	Strict Template
Price Drop / Inventory Alert	MED	MED	Template + Slots
Social Re-engagement	HIGH	HIGH	LLM Generation
Content Discovery	HIGH	HIGH	LLM Generation
Abandoned Cart Recovery	HIGH	MED	LLM + Guardrails

THE MARGINAL VALUE QUESTION

PushGen (+14.08%) and e-commerce (+12.5%) gains occurred where baselines were **underdeveloped**. DoorDash's +1.8% reflects a system where substantial upstream optimization had already occurred. LLMs do not become ineffective in mature systems, but their deployment requires more careful measurement.

The Unified Architectural Framework



! Key Failure Mode Tradeoffs

🛡️ Guardrail Throttling

Strict factuality and policy filters prevent brand damage but can **block too many valid messages**, forcing the system to fall back to generic templates and reducing overall campaign ROI.

🎯 Exposure Bias

The reward model only learns from messages that were actually sent. Over time, it becomes **blind to novel linguistic patterns**, leading to a feedback loop that reinforces repetitive, formulaic copy.

⌚ Latency Overhead

LLM generation adds significant latency. If generation takes too long, the message may **miss the optimal send-time window** identified by the contextual bandit, negating the value of the personalized copy.

Evaluation Must Go Far Beyond Short-Term CTR

Domain Generalizability Matrix			
Dimension	Social Media	Food Delivery	E-Commerce
"High-Value" User Definition	High-DAU user with strong social graph; influencer or power user	Repeat orderer with high AOV; lapsed user with reactivation potential	High-CLV customer; abandoned-cart user with purchase intent
Primary Success Metric	Engagement Rate (likes, shares, comments) Session re-entry	Order Conversion Rate GMV per notification Reactivation rate	Click-to-Purchase CVR Revenue per notification Cart recovery rate
Key Contextual Signals	Social activity feed Friend/follower events Content freshness	Time of day (meal windows) Order history & recency Location & weather	Browse & cart history Price drop / stock alerts Seasonal promotions
LLM Role in Pipeline	Personalise social recap & trending content summaries	Generate contextual urgency copy (e.g., lunch offers)	Produce persuasive product descriptions & recovery messages
Persuasion Tolerance	Moderate — users expect social updates, not hard selling	High during meal windows; low outside peak hours	High for deal/discount frames; low for generic promotions
Dominant Failure Mode	Notification fatigue from over-frequency; tone mismatch	False urgency (dark pattern risk); stale context	Hallucinated prices or stock levels; reward hacking
Cross-Domain Transferability	GNM audience selection transfers; LLM copy style does not	STO bandit transfers; urgency framing is domain-specific	RAG grounding transfers; CLV segmentation transfers broadly

Shaded left bars indicate row dimension category. Green cells = transferable across domains, white cells = domain-specific.

🎯 Tier 1: Primary Metrics

Short-term engagement: Click-Through Rate (CTR), Conversion Rate (CVR), and immediate revenue. While easy to measure, optimizing solely for these leads to clickbait and reward hacking.

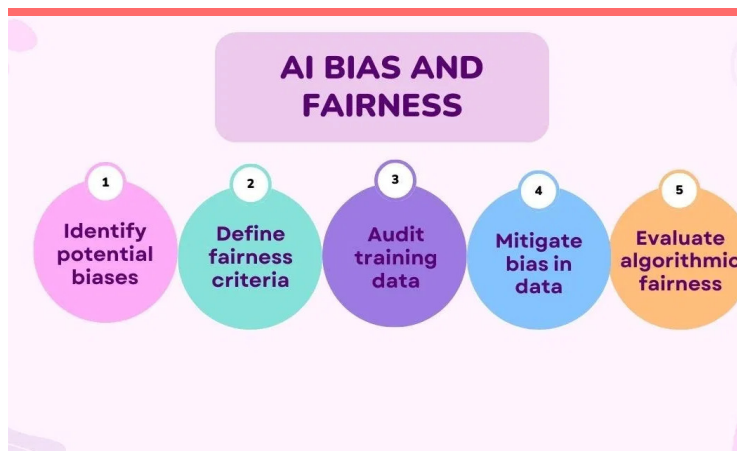
🛡️ Tier 2: Guardrail Metrics

User autonomy & safety: Opt-out rates, notification permission revocations, and negative feedback. These must be strictly bounded during A/B testing.

📈 Tier 3: Long-Horizon

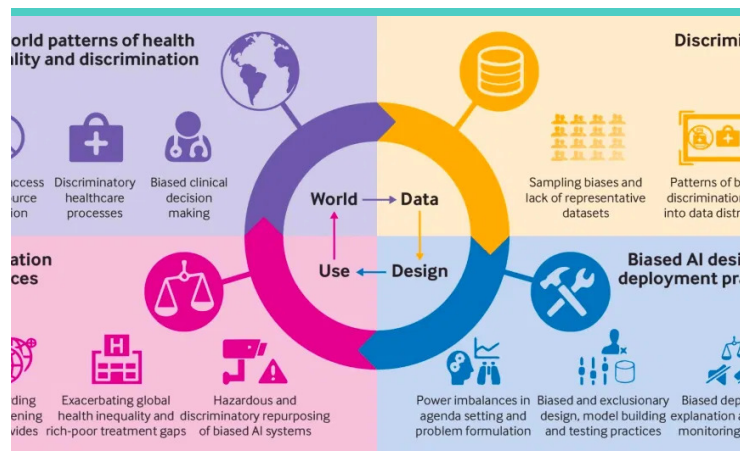
Ecosystem health: Customer Lifetime Value (CLV), 90-day retention, and platform uninstall rates. These capture the delayed effects of notification fatigue and trust degradation.

Ethical Boundaries: Personalization vs. Manipulation



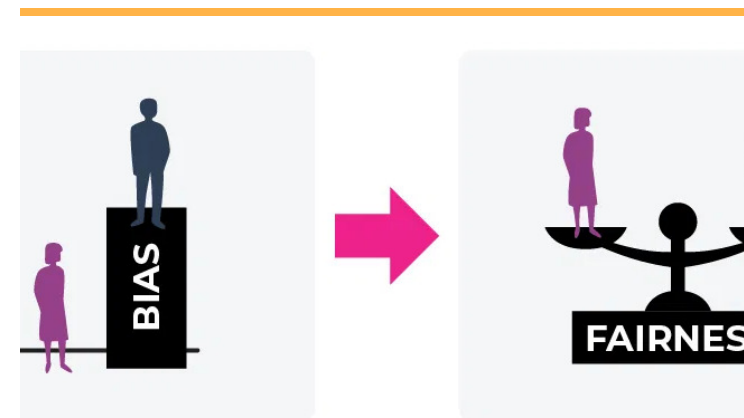
Persuasive Asymmetry

Platforms possess vast informational advantages over users. LLMs can weaponize this asymmetry by generating hyper-personalized "**dark patterns**" (e.g., false urgency or scarcity) that cross the line from helpful nudges into coercive manipulation.



Differential Susceptibility

Not all users respond equally to persuasive text. Reward models optimized purely for CTR may inadvertently learn to target **vulnerable demographics** with aggressive messaging, raising severe fairness and algorithmic bias concerns.



Regulatory Compliance

Under **GDPR Article 22** and **DSA Article 40**, automated systems with significant user impact require strict auditability. Black-box LLM generation complicates compliance, necessitating transparent guardrails and human-in-the-loop oversight.

Lifts Range from +1% to +14.5% — Baseline Sophistication Drives the Variance

+1% to
+14.5%

Reported CTR improvement
over static templates

The Baseline Effect

The variance in reported lifts is driven primarily by **baseline sophistication**. Systems replacing static templates see +8–14.5%; systems competing against mature slot-filling see +1–2.5%.

These figures span heterogeneous systems, baselines, and evaluation conditions. They should not be treated as directly comparable.

PRODUCTION EVIDENCE — MAPPED TO QUALITY DIMENSIONS

SYSTEM	DOMAIN	PRIMARY QUALITY DIMS	KEY RESULT
PushGen	Social Media	Freshness, Novelty Handling	+14.08% CTR
Instagram Diversity	Social Media	Freshness, Novelty Handling	Better CTR, lower volume
E-Commerce LLM	Retail	Actionability, Persuasive Framing	+12.5% CTR, +8.3% CVR
DoorDash	Food Delivery	Contextual Relevance, Clarity	+1.8% engagement

6×

The six quality dimensions **compound**: a message improved on multiple dimensions simultaneously outperforms single-dimension improvements. This compounding explains why the largest gains (+12–14%) exceed what any single dimension predicts.

Threats to Validity and Future Research Directions

⚠️ Five Threats to Validity



Publication Bias

Positive results are heavily overrepresented; failed experiments and negative A/B tests are rarely published.



Temporal Degradation

Technical constraints (e.g., inference latency) reported in 2024–2025 may be resolved by hardware advances in 2026.



Domain Representation

Heavy focus on Social Media and E-Commerce; findings may not generalize to healthcare or financial alerts.



Reproducibility

Most important industrial systems depend on proprietary data and internal frameworks, lacking open benchmarks.



Framework Limitations NEW IN V3

The pipeline trades latency/compute for quality. Performance depends on the Budget-Aware Router's CLV accuracy and counterfactual logging quality.



Future Research Priorities



Multimodal Generation

Jointly generate personalized copy, images, and rich media layouts tailored to individual user preferences.



Causal Inference Methods

Robust frameworks to bridge the gap between offline reward model evaluation and online A/B test performance.



Federated Learning

Training personalization models on edge devices to ensure privacy compliance while maintaining quality.



Direct Preference Optimization (DPO)

Exploring DPO as a more stable alternative to RLHF in notification systems.



Cross-Domain Transfer

How stylistic priors learned in one domain (e.g., social media) can transfer to another (e.g., e-commerce).

Five Key Takeaways from v3

01 **Message quality is an independent, underinvested lever.**

Modern stacks optimize who, when, and what near-perfectly. The how — message composition — is the widest remaining gap and the one where LLMs add differentiated value.

02 **LLMs improve message quality along six concrete dimensions.**

Gains concentrate along freshness, contextual relevance, and actionability — precisely where templates are structurally weakest. Improvements compound across dimensions.

03 **Attribute gains to the message layer, not the full pipeline.**

LLMs should be evaluated against message-layer baselines. Observed CTR lifts (+1% to +14.5%) span heterogeneous systems and should not be treated as directly comparable.

04 **The negative case is as important as the affirmative one.**

The binding-constraint framework specifies when not to deploy LLMs. Transactional and low-variance alerts are served better by templates — LLMs add cost without quality.

05 **Evaluation, ethics, and governance are non-negotiable.**

Guardrail metrics (opt-outs, mutes), long-horizon health metrics, fairness audits, and factuality guards are core production requirements, not optional accessories.

"The binding-constraint framework offers practitioners a principled basis for making deployment decisions, and for deciding when not to deploy LLMs at all. That negative case is as important as the affirmative one."

— Agrawal (2026), Section 12

FUTURE RESEARCH PRIORITIES

Progress requires **stronger experimental design** with component-isolated evaluation, better off-policy learning infrastructure, more rigorous **fairness auditing**, and open benchmarks that disentangle the contribution of generation from the broader recommender stack.

The six-dimension message quality framework proposed here offers a **starting point** for that more rigorous conversation.